

Q-JEPA Benchmarking Plan: Baselines and Published Results

Companion document

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1 Tier 1: Gold Standard Benchmark

[1] Bowles, Ahmed, Schuld (arXiv:2403.07059, 2024). “Better than classical? The subtle art of benchmarking quantum machine learning models.” Code: <https://github.com/XanaduAI/qml-benchmarks>.

12 quantum models $\times$ 6 tasks $\times$ 160 datasets. **Findings:** (i) classical ML outperforms quantum out-of-the-box; (ii) removing entanglement often helps; (iii) no quantum model consistently wins. **Why mandatory:** community reality check — any QML paper must use their protocol.

2 Tier 2: Closest QML Methods

Method	Key Result	Citation	Why Compare to Q-JEPA
Quantum Kernel SVM (QSVM)	5-qubit IBM demo; 0.779 acc on consumer classification [13]; competitive on Bowles hidden manifold	[2] Havlíček et al., <i>Nature</i> 567 , 209 (2019)	Q-JEPA with $V = \text{id}$ IS a QSVM. Our direct baseline.
Projected Quantum Kernel (PQK)	Avoids exponential concentration via local RDMS; competitive with classical for $n \leq 15$ [13]	[3] Huang et al., <i>Nat. Comms.</i> 12 , 2631 (2021)	Closest to our local purity approach. PQK builds kernel from local RDMS; we regularize with them.
Equivariant QNN	S_n -equivariant: no BP, poly generalization [4]; rotationally equivariant: “dramatically outperforms generic” [6]; $\mathbb{Z}_2^2$: outperforms with 50% fewer params [14]	[4] Larocca et al., <i>PRX Quantum</i> 3 , 030341 (2022); [5] Ragone et al., <i>Nat. Comms.</i> 15 , 7172 (2024); [6] Sauvage et al., <i>PRX Quantum</i> 5 , 030320 (2024)	Tests our Lie algebra theory: equivariance $\Rightarrow$ restricted $\dim(\mathfrak{g}) \Rightarrow$ trainability. Can purity reg match without knowing G ?
QCNN (local cost)	BP-free by architecture; 98.7% acc on image classification (up from 52.3% baseline)	[7] Pesah et al., <i>PRX</i> 11 , 041011 (2021); [8] Chen et al., arXiv:2603.11131 (2026)	Architectural locality vs. our regularization locality. Which BP fix wins?
Data re-uploading	Single qubit is universal classifier; $\sim 97\%$ on Iris; standard baseline in [1]	[9] Pérez-Salinas et al., <i>Quantum</i> 4 , 226 (2020)	Simplest QML model. If Q-JEPA can’t beat this, the framework isn’t justified.

Foundational results our theory builds on: [10] Cerezo et al., *Nat. Comms.* **12**, 1791 (2021): local cost $\Rightarrow \Omega(1/\text{poly})$ gradients; global $\Rightarrow O(1/4^n)$. [11] Thanasilp et al., *Nat. Comms.* **15**, 5200 (2024): exponential kernel concentration for expressive circuits. [12] Schuld, arXiv:2101.11020 (2021): all supervised QML models are kernel methods.

Method	Library	Rationale
RBF-SVM	scikit-learn	Classical kernel. No kernel advantage if QSVM $\leq$ RBF-SVM.
Gradient-Boosted Trees	XGBoost	Strongest classical on tabular data. Used in [1].
MLP (2-layer)	scikit-learn	Neural net at comparable parameter count.
Linear SVM	scikit-learn	Lower bound. If quantum $\leq$ linear, stop.

Dataset	Source	N	d	Rationale
Hidden Manifold	Bowles [1]	tunable	tunable	Probes quantum geometric advantage
Two Curves	Bowles [1]	200	2	Quantum-friendly nonlinear boundary
MNIST (0 vs 1)	Standard	1000	n (PCA)	Image benchmark
Fashion-MNIST	Standard	1000	n (PCA)	Harder than MNIST
Iris	Standard	150	4	Sanity check

3 Tier 3: Classical Baselines

4 Datasets and Protocol

Protocol (following [1]): 80/20 stratified split; 5-fold CV for hyperparameters; report accuracy, F1, AUC-ROC (mean $\pm$ std, 10 seeds); scale $n = 4, 6, 8, 10$ for MNIST/Fashion-MNIST; report $\text{Var}_\theta[\partial L/\partial \theta_0]$ over 200 initializations vs n (the key diagnostic).

5 Experimental Design

Comparison	Hypothesis Tested
Q-JEPA+reg vs QSVM	Does predictor V + purity regularizer add value over plain kernel?
Q-JEPA+reg vs PQK	Regularization vs local kernel construction?
Q-JEPA+reg vs Q-JEPA (no reg)	Ablation: does regularizer prevent BP at $n \geq 8$?
Q-JEPA vs QCNN	Regularization vs architectural locality?
Q-JEPA vs equivariant QNN	Regularization vs symmetry exploitation?
All quantum vs RBF-SVM	Does <i>any</i> quantum model beat the classical kernel?
Gradient variance vs n	Money plot: Q-JEPA flat at ~ 0.017 while QSVM decays as $\sim 0.36^n$

6 Expected Results

At $n \leq 10$: Q-JEPA accuracy will likely *match* QSVM/PQK and be matched by RBF-SVM (standard finding [1, 13]). The separation is in **gradient variance**: our data shows local gradient flat at ≈ 0.017 vs global $\sim 0.36^n$, and the ablation (with/without regularizer) should show training collapse at $n \geq 8$ without it. On symmetric data, equivariant QNNs may win; on generic data, Q-JEPA is architecture-agnostic. QCNNs (98.7% [8]) may win on images due to convolutional bias; Q-JEPA’s advantage is generality.

The honest conclusion: “We prove gradient variance stays positive, demonstrate it empirically, and provide the M_2 framework that *predicts* trainability. On standard benchmarks, performance is competitive but not superior to classical at current scales. The contribution is the theoretical explanation of when and why quantum methods work.”

References

- [1] J. Bowles, S. Ahmed, M. Schuld, arXiv:2403.07059 (2024).
- [2] V. Havlíček et al., *Nature* **567**, 209 (2019).
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- [13] R. Kübler et al., *Quantum Mach. Intell.* (2025).
- [14] T. Nguyen et al., *Axioms* **13**(3), 188 (2024).